

Reconsidering Money and Inflation: Groundwork for an Alternative Macro Basis

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Abstract

In this essay I explore the definition of money, proposing a strict binary test: *Money in a given market is any good which is directly used as a medium of exchange, or can be converted into a medium of exchange without requiring the transfer of another agent's medium of exchange.* This test is used to classify the most common money-like instruments. For example, highly liquid assets such as Money Market Funds are not money, and internal ledgers of non-depository institutions (such as paypal, clearinghouses, broker-dealers) create money. Two monetary aggregation weighting schemes are critiqued, resulting in the conclusion that simple summation is superior. I then move on to inflation, proposing a formulaic definition based on the equation of exchange, under strong stable money velocity assumptions. The case is made for including investment assets (financial assets and land) to produce a truly general purpose inflation measure. Finally, for each major investment asset sector, I propose a starting place for integration with standard indices by measuring quantity changes directly and modelling quality changes.

Contents

1	Money	1
1.1	Definition	1
1.2	Forms of Money?	3
1.2.1	Central Bank Liabilities	3
1.2.2	Checking & Savings Deposits	4
1.2.3	Time Deposits	5
1.2.4	Treasury Bills	5
1.2.5	Repo Assets	7
1.2.6	Money Market Funds	7
1.2.7	Interbank Deposits	8
1.2.8	Internal Ledgers	8
1.2.9	CHIPS	11
1.2.10	On Simple Sum v.s Component Weighting	12
2	Inflation	23
2.1	Unit Elasticity of Diverse Aggregates	26
2.2	General Inflation Definition	27
2.3	General Inflation Metrics	30
2.3.1	Consumer Durables, Investment Durables, Inventories	34
2.3.2	Consumer Nondurables and Non-OER Services	34
2.3.3	Land	35
2.3.4	Share of Franchise Value	36
2.3.5	Debt	37
2.3.6	Other	38
2.3.7	Financial Asset Quality Adjustment	38
2.4	Implementation Groundwork	41
3	Conclusion	44

1 Money

It seems unlikely a firm understanding of macroeconomics is possible without a firm understanding of the fundamental properties of money, which would seem to require a single, authoritative, objective definition. Yet conventional macroeconomics swims in a soup of aggregates from M0, M1, M2, M3, M4, Divisia aggregates, etc., each differing in what concepts are deemed to be most important to capture under the term “money” (maybe it’s all just contextually relative), and/or where to place relevant cutoffs. Some even take the view that our highly complex modern financial system has made the concept of money obsolete (Hoover, 1988), and New Keynesian theories don’t even include money in their models (Beck and Wieland, 2010, p. 5). However, increased complexity only makes a firm understanding more critical.

1.1 Definition

In this essay I propose a single universal definition of money which I contend paints the clearest picture from a macro perspective, avoiding double and under counting: *Money in a given market is any good which is directly used as a medium of exchange, or can be converted into a medium of exchange without requiring the transfer of another agent’s medium of exchange.* Other than the fact that I ignore the *unit of account* and *store of value* conditions (which are second order downstream properties arising from the *medium of exchange* property), the key differentiator from (implicit) mainstream definitions underlying monetary aggregates, is the assertion that the most fundamental property of money is not degree of liquidity, an amorphous property which is difficult to formalize (although Divisia aggregates have a measurable micro-founded method to quantify share of money services per component money good, using interest rates (Barnett, 1980, pp. 19–20), of which liquidity is arguably the most critical aspect by far) but rather the mechanism of payments. Liquidity is a downstream byproduct of more fundamental facts about payment mechanics, and other external market factors not unique to money.

This definition is conditional on a given market, the most straightforward implication of which is the differentiation between what counts as money in different foreign markets, and while it’s true that in US market ¹ the Swiss Franc doesn’t function as money since it needs to be sold into dollars before being used to purchase anything,

¹I will focus on the US economy for simplicity, but the same arguments apply to any similar fiat system.

the more consequential difference, from a broader perspective above the boundaries of national currencies, is between central bank reserves and private depository institution's ² deposit liabilities. For example, reserves are a medium of exchange for the interbank market, while deposits are a medium of exchange for the nonbank market, but reserves are not a medium of exchange for the nonbank market. The US nonbank market is far larger and more relevant to the US economy. Many of the reasons why the interbank market is relevant to the US economy has to do with it's indirect effect through financing for the nonbank market. Thus when we ask what is money for the US economy, we should ask what is money for the nonbank market in the US.

While the first sufficient condition of being a medium of exchange is obvious, why should goods that are not used as mediums of exchange in their current state be included? Because, the definition should capture not just market states in the current moment, their current manifested monetary kinetic energy, but also unmanifested potential energy that could be released at any moment with relatively little friction (as opposed to the creation of new money by bank lending). The concept of money would be too limited and unreliable for use in longer time horizons if it only included goods that are mediums of exchange in their current state. Additionally, the definition should not be too sensitive to the sweeping discipline of market participants. For example, just because sellers keep profits from a sale in their checking accounts for longer than they usually would before transferring them to savings, doesn't mean the supply of money has increased in any meaningful way. This would clearly be far too sensitive to trivial financial sweeping program innovations.

The importance of considering goods with potential monetary energy that can be converted into money is accepted by users of conventional simple sum aggregates (broader aggregates include Money Market Funds (MMF) for example), but the speed and ease of that conversion is incorrectly stressed, while the mechanical nature of the conversion is not. While this may make sense at the micro level, at the macro level it results in double counting, and no clear consensus on how to measure "liquidity", and where thresholds should be set.

The real fundamental distinction between a non-monetary asset and money is whether or not the asset has to be *sold* before it is transformed into the most liquid form, at which point it can be used to transform into any other good on the market. If the asset has to be sold to transform, that means it's subject to market conditions.

²Private depository institutions will be called "banks" even though the category includes more than just commercial banks.

If it has to be sold to become a medium of exchange, that means it's not money itself. For the individual, the asset 'becomes' money only when an external buyer is found (from the macro perspective it's just an exchange, not a literal transformation) and provides their actual money in exchange for the good. This is key because a change in the supply/demand of a non-monetary good directly affects only the price of that good, even though there may be secondary effects. A change in supply/demand of actual money however affects prices of all goods directly, because it's directly exchangeable for all goods. Evolutions in the financial system resulting in new supplies of highly liquid non monetary assets do not change this fundamental fact, even though they can reroute the direction and velocity of money flows, connecting pools of cash providers and cash borrowers/spenders that were previously unconnected. Analysis becomes more clear if these money flow evolutions are addressed explicitly as a separate phenomena rather than muddying the definition of money itself. If a hurricane sweeps through a landscape, resulting in flooding, which changes the topography of a water shed, do we change the definition of water, or do we say the river's velocity and path have changed?

1.2 Forms of Money?

In this section I will lay out exactly which financial instruments pass my definition of money and why, including: physical currency, reserves, foreign official deposits, checking deposits, savings deposits, non-negotiable CDs (Certificates of Deposit), negotiable CDs, treasury bills, repo assets, MMFs, interbank deposits, various internal ledgers, and Clearing House Interbank Payments System (CHIPS) "credits".

1.2.1 Central Bank Liabilities

Central bank reserves and central bank foreign official deposits are money for the interbank market, because they're the actual settlement layer, but not for the nonbank market, which is the much more relevant market to consider to understand the impacts of money supply on prices and the broad economy, which includes both the "real economy" as well as financial markets, both in which the vast majority of transactions happen between nonbank counterparties. Physical currency is theoretically money for both markets, but practically only for the nonbank market, as banks typically settle in reserves since moving physical cash around is extremely inefficient, relative to CHIPS/fedwire settlement. Thus physical currency behaves more like a non-monetary asset that banks buy and sell to each other in order to be able to provide

physical currency services to their customers.

It's not correct to count other central bank liability classifications, such as standing repo facility liabilities, as money, in addition to reserves/foreign official deposits/physical currency, because those are all settled in the aforementioned settlement layers. The point is to define the liquid itself, not its flow, avoiding double counting. From this point on, I will be focused on money for the nonbank market, unless otherwise stated.

There is a very important caveat to keep in mind when counting reserve liabilities: official Federal Reserve series such as the WRBWFRBL use end of day levels; they do not account for intraday fluctuations ([Board of Governors of the Federal Reserve System, 2012](#)). This is relevant because private banks depended very heavily on intraday credit from the federal reserve (daylight overdrafts) prior to the GFC ([Board of Governors of the Federal Reserve System, 2026a](#)), which represent enormous expansions of average reserve levels which are not tracked by series such as WRBWFRBL.

1.2.2 Checking & Savings Deposits

Checking deposits are clearly money. Funds can also be sent directly from savings to savings, savings to checking, checking to savings to settle a transaction, but is typically moved through the checking to checking channel ([The Federal Reserve System, 2016](#), p. 1). But even if it was impossible to send/recieve money directly from savings, it would still qualify as money because it “can be converted into a medium of exchange without requiring the transfer of another agent’s medium of exchange”, i.e. it can be transferred to checking without requiring a market transaction. This is not to say there are no meaningful liquidity differences between checking and savings, but rather that this isn’t the defining feature of money, as explained earlier. Additionally, there are labeling complications such as savings accounts with very little withdrawal restrictions, and unique checking accounts such as Interest Checking, High-Yield Checking, Rewards Checking, and NOW accounts, which offer yield and can act more like a savings account ([Board of Governors of the Federal Reserve System, 2011](#), pp. 1–2). A better distinction for liquidity purposes is yield-bearing vs non-yield-bearing, but this still doesn’t address the confounding variable of sweeping behavior. Even after accurately separating high velocity money from low velocity money (fixed interval balance sheet snapshots could be correlated with window dressing and sweeping behavior and thus are insufficient), a definition of money with only high velocity accounts would be incomplete: Suppose there are 10 dollars circulating in the entire nonbank market in high velocity accounts at t_0 . At t_1 , a private bank

credits 90 dollars to an individual's low velocity account. They do not spend it immediately, but intend to eventually. Is it correct to say the money supply at $t1$ is still only 10 dollars? If the definition of money is focused on accurately describing liquidity with relation to only immediate present, then it may be close enough to say the money supply has not increased at $t1$. However if the definition of money is to have broader implications for longer time horizons, then it's wrong to say the supply of money has not increased at $t1$ even if it's currently sitting in a low velocity account. That large savings account is like a dammed lake of stored "potential energy" which could release high velocity liquid at any moment, since there is typically not that much friction in transferring deposits from checking to savings, and also it could even be spent directly from savings. The definition of money is like the definition of a liquid, not the description of how that liquid flows, and a dammed lake surely meets the criteria of a liquid even though it doesn't have high velocity.

1.2.3 Time Deposits

Non-negotiable CDs are essentially even lower velocity forms of savings accounts, and are money for the same mechanical reason that they can be converted into a medium of exchange without requiring the transfer of another agent's medium of exchange. This will typically happen when the contract's lock up period ends, but can happen early, with fees applied (Board of Governors of the Federal Reserve System, 2011, pp. 2–3). Negotiable CDs are unique in that they can be sold in the market (Board of Governors of the Federal Reserve System, 2011, pp. 2–3), acting more like a non-monetary good. Yet the fact that they can be sold second-hand does not change the fact that they can also be converted into a checking account without requiring a market transaction, and by the same token a checking account can be converted into a CD without a market transaction taking place.

Mediums of exchange can move in and out of CDs, without marketplace transfers of existing goods. Thus if large pools of money decide they want to move into CDs, including negotiable ones, a money definition/aggregate without non-negotiable and/or negotiable CDs will incorrectly say money supply has decreased, when in reality it has merely been converted into a less liquid form, an important distinction.

1.2.4 Treasury Bills

As it has been alluded to, bills are generally not money because they are not directly used as a medium of exchange (except for Delivery versus Delivery (Committee

on Payment and Settlement Systems and Technical Committee of the International Organization of Securities Commissions, 2012, p. 76), which are rare edge cases International Capital Market Association and ICMA European Repo and Collateral Council, 2018, p. 67; Banque de France, 2018, p. 213), nor can they be converted into one without the transfer of another agent's medium of exchange, either when they are sold on the market, or when held until maturation. Even though, in the held to maturity case, the Treasury pays the treasury bill holder's bank in reserves, which are not money for the nonbank market, the Treasury obtained these reserves through taxes largely levied on the nonbank sector. This means the Treasury is just a temporary-holder-of-funds-middleman between the taxpayers and bill holders, whereby a medium of exchange (checking) is required to be transferred from the taxpayers to the billholders (in the held to maturation case) in order for the bill to be converted into a medium of exchange. The other transformation option, sale, is also a transfer of a medium of exchange from the buyer to the seller. The caveat to both of these cases is when the nonbank market and interbank market interact, specifically when the funds-providing counterparty is a bank and the funds receiver is a nonbank, i.e. when the taxpayer is a bank and the maturing bill holder is a nonbank, or the bill buyer is a bank (private bank or central bank) and the seller is a nonbank. The extent to which both of these occur is the extent to which bills are money, because these are transformation cases in which a non-monetary good, reserves, becomes a monetary good, private bank deposits, rather than a transfer of ownership of private bank deposits.

The reason why payment mechanics are so critical can be seen through the lens of extreme supply changes and their impacts on prices. If the supply of gold increases tenfold, the prices of other goods will not be majorly impacted in a direct way, which, ultimately, is because gold is not used as medium of exchange, and consequently, other goods are not priced in gold. What *will* be dramatically impacted is the price of gold itself. In a similar vein, if the Treasury increases its bill issuance tenfold, the prices of all dollar denominated goods will not increase proportionally, which they should if bills were money (roughly, based on equation of exchange). They would likely actually do the reverse, at least temporarily, falling in price as primary dealers are forced to fire sale their other assets and take on massive debt in order to absorb new treasuries, placing massive demand pressure on actual money: bank deposits. Bank balance sheets would be strained and they would likely be forced to cut down on other deposit expansion, bringing prices down until the Treasury floods the money

it borrowed back into the economy. Because bills are not a medium of exchange, and generally cannot be converted into one without a market transaction, supply and demand changes in treasuries primarily affect just the price of the bills themselves, not every price in the nonbank market, making it incorrect to view them as money. The high liquidity of bills does not change this fundamental fact. Similar logic can be applied to highly rated corporate paper or Mortgage Backed Securities.

Money is water and bills are rivers that influence the flow and velocity of that money, but are not money itself. Just like if you wanted to count the volume of all the water in a watershed, you would simply count the volume of water itself, including lakes, ponds, and rivers, and would not add to that total all the water flowing through rivers again; that would be double counting.

1.2.5 Repo Assets

The same argument for why bills are not money applies to all debt assets where the borrower is a nonbank, including repo. Again, the caveat is at the intersection of nonbank and interbank markets, where the lender is a nonbank and the borrower is a bank (including the Fed through its ON RRP facility). In this case the nonbank's repo asset is essentially a deposit under a different label. While it's not a medium of exchange itself, it can be converted into one without the transfer of another agent's medium of exchange (reserves do not count for the nonbank market). When the bank repays its repo liability, it doesn't hand over its own private bank deposits, rather it simply credits the lender with new deposits.

1.2.6 Money Market Funds

MMFs are trickier because they're an additional layer of claims on a diverse set of assets, but taken case by case, the same logic applies. When deposits are transferred to a MMF, that fund is a nonbank and holds those deposits in its own bank account/s. Thus if all private bank deposits are counted, and the portion of MMF shares which are backed by deposits in the MMFs bank accounts are added, the result is double counting. The portion of inflows the MMF spends on bills purchased from nonbanks, or lends in repo to nonbanks, are also already counted by considering all private bank deposits, and counting the portion of MMF funds backed by these non monetary assets in addition to counting deposits is also double counting. Again, however, the two caveats are: 1) the portion of MMF assets held in bills that will be sold to banks, and 2) the portion of MMF assets in repo to banks. When constructing an aggregate,

these caveats should already be addressed when accounting for the bills and repo goods themselves.

1.2.7 Interbank Deposits

For the interbank market, interbank deposits are money because they're accepted among banks as a medium of exchange (among "lower tier" banks that use the same correspondent bank) ([Committee on Payment and Settlement Systems, 2003](#), p. 3). They're not money for the nonbank market for the same reason reserves aren't; non-bank participants can't hold interbank deposits.

1.2.8 Internal Ledgers

The first type of internal ledger to consider is a nonbank wallet ledger keeper such as paypal (stablecoins also follow this model). Paypal/Venmo/Cashapp credits are accepted as a medium of exchange to a limited extent, which qualifies them as money. Again, The potential issue with this classification is double counting: if the nonbank wallet ledger keeper simply parked customer funds in idle deposits, then inclusion of these credits into an aggregate would be double counted by the keeper's idle deposits. However, typically the keeper uses customer funds to purchase safe short term assets such as treasury bills/repo/MMFs ([Speights, 2023](#)), in which case the effective supply of money actively being used in the marketplace is greater than the sum total of only wallet ledger credits, and would need to be counted along with customer deposits.

The second major type of internal ledgers are broker-dealer free credits. These can either be credited to a cash account account or a margin account. In either case, they're a medium of exchange on top of deposits because they can be issued and spent as payments to settle market transactions, without the transfer of deposits. The customer can sell an asset to the broker-dealer, who finds a buyer at the other end (at some point), and can credit the seller rather than sending deposits. These credits could be mirrored by margin debits in the broker-dealer-intermediated-buyer's margin account, or not. If the broker-dealer has issued more credits than debits, a mirrored deposit portion must be kept in a "special reserve bank account", which can be invested in qualified securities ([Legal Information Institute, Cornell Law School, 2026](#)). The credited seller can then spend those funds to buy assets from the broker-dealer, which is ultimately just an intermediary for many other end sellers. It does not matter that the credits are not directly transferred from seller to buyer, what

matters is that they can effectively be used as payments for transactions in place of deposit transfers, and thus they are mediums of exchange.

The third major type of internal ledgers are central counterparty (CCP) clearing house net credits/debits, which are settlement obligations organized by the netting ledger of a clearinghouse. CCPs are more similar to broker-dealers than positive-only wallet/broker-dealer ledgers, as they intermediate credits and debits between buyers and sellers, creating money outside of deposits to the extent that these unsettled credits exist. There are two major differences from broker-dealer credits: 1) credits are routinely destroyed at the end of a netting cycle, 2) credits do not exceed debits and vice versa. Broadly speaking, CCPs create credits by the following process: transactions between CCP members are executed on an exchange and sent to the CCP which begins a netting cycle, during which the CCP determines who are the debtors and who are the creditors, creating mirrored debt liabilities and assets. When the netting cycle is complete, credits and debits “destroy” each other as final settlement occurs through deposit transfers. Legally, the CCP is responsible for credits, just like wallet ledger keepers and broker-dealers (counterparty risk reduction is one of the major benefits of CCPs), but economically, it’s redistributing those credit liabilities to the various debtors. Yet just like other credits, the net credits displayed to members accounts during the netting cycle fundamentally function in just the same way, meaning they can be used as payments for transactions. The fact that a CCP member may not explicitly think they are receiving net credits as payments for assets or spending those net credits to purchase assets later in the same netting cycle is economically irrelevant, as well as the fact that those credits aren’t explicitly transferred from buyer to seller. What matters is that these credits are being used as settlement layers outside of deposits, while they exist.

If at any point in time t every transaction up to t in the current netting cycle has perfectly netted out, then economically it’s equivalent to an instantaneous settlement situation where each party sent each other deposits for each individual transaction. In this case, there’s no excess money in the system in a way that is not represented by deposits. If some more transactions occur between t and $t + 1$ (not the next day but future point in time in the same netting cycle), some portion will be perfectly netted out, and again for this portion of transactions, it’s as if existing deposits were instantaneously transferred between all the parties. However the resulting net credits and debits at $t + 1$ represent transactions where it was as if the debtor/buyer created a new transaction medium, new money, and sent those net credits to the sellers as

payment for their purchase, whilst also creating a debt obligation for themselves. From $t + 1$ to $t + 2$ more transactions will happen and again, some portion will be fully netted, while the remaining transaction portion involving the net creditor from $t+1$ could result in their net credit position being reduced, which is economically equivalent to sending net credit money as payment to the net creditors of $t + 2$.

It's crucial to notice that net debits are a form of debt which is fundamentally different from any other kind of debt: the lender, which could be conceptualized as the CCP or net creditor (economically irrelevant), did not have to transfer any funds. Normal debt represents transfer of money, from the lender to the borrower. In this case no deposits, no money, are required to be sent from a lender to a borrower, the CCP can simply allow the borrower to create funds *ex nihilo*. It is as if these newly created net credits are sent from the borrower/buyer to the seller, simultaneously creating a debt obligation for the borrower/buyer. Of course the borrower is limited in how much money they can create as they will eventually have to transfer an equal amount of deposits when the netting cycle ends (and will have account specific limits), effectively destroying those net credits (along with the net debits) as they're replaced by existing deposits. This is the economic benefit of not having to prefund: it allows CCP members to "print" temporary money. It may be the case that members *are* actually effectively prefunded if they set aside earmarked deposits covering their net debits at the moment of trade execution, but they do not have to be. Prefunding behavior should be captured by measures of money velocity, not supply. More on this in [subsubsection 1.2.10](#).

A practical sidenote to consider when constructing an aggregate: CCP net credits are unique in that they expand and contract with much higher frequency and magnitude (relative to their own average level) compared to other forms of money. This can be dealt with by calculating a smoothed moving average level when adding them to the monetary aggregate.

The key properties which all three types of internal ledger credits share are: 1) borrowing through balance sheet expansion / credit creation rather than merely transferring existing funds from a lender. 2) credits are accepted as a medium of exchange / settlement layer. 3) non idleness of deposits backing credits. The combination of these three characteristics result in money creation for the nonbank market, beyond deposits and cash. Wallet ledgers / stablecoins satisfy 1): the credit issuer is essentially "borrowing" deposits from the customer, which they are liable to pay back on demand, while the customer/"lender" does not have to give up any money.

Their deposits simply become credits, which satisfy 2). The issuers satisfy 3) as they use their borrowed deposits to purchase money market instruments, treasuries, etc, keeping those funds circulating as opposed to sitting idle. Broker-dealers satisfy 1) because they either intermediate margin borrowing or “borrow” deposits from net credit holders themselves, but without requiring creditors to give up money, as again their deposits simply change form into credits, which satisfy 2). 3) is satisfied because broadly speaking, intermediated margin borrowers do not keep idle funds to back their debt as that would defeat the purpose of borrowing on margin. When credits exceed margin debits, broker-dealers are required to keep a portion of funds in the “special reserve bank account”, but these funds are not required to sit idle in bank accounts. CCP net credits satisfy 1) and 2) as explained in the previous paragraph, and satisfy 3) to the extent that clearinghouse members do not prefund.

1.2.9 CHIPS

The CHIPS Network, owned and operated by The Clearing House, is a clearing system for interbank payments, with no direct money supply implications for the nonbank market. CHIPS does however expand the supply of interbank money. CHIPS operates in the following way: participants prefund their accounts with reserves at 9pm, and CHIPS credits participants with an internal ledger account balance equal to the prefunds. During the 21-hour processing day, participants queue payments. Those queued payments are released in multiple batches throughout the day, settling on the CHIPS internal ledger, in such a way that ensures no negative balances. Participants add supplemental funding to their accounts throughout the day. Between 6:00-6:30pm, participants are required to fund their accounts so that remaining queued payments that would otherwise result in negative balances can be released, settling on the internal ledger, which is then finally settled with reserves. Unfunded payments expire, with no liability for CHIPS ([The Clearing House Payments Company L.L.C., 2025](#)). Central liability is the key difference between CHIPS and securities clearing, yet economically, at a high level, they are very similar. This is not immediately apparent, because of the fact that CHIPS does not incur any central liability. CHIPS cannot novate funds, because individual payments must be reversible if funding is not met by the end of the 21-hour cycle. However the same net credit/debit measure at any point in time can be reconstructed given all queued payments at that time, which would be equal to the gross dollar value of all negative balances if all queued payments were optimally netted. While it is true that within the CHIPS system itself, participants

only see their current CHIPS-ledger-settled balance excluding queued payments, they should be fully aware of their current balance including queued payments, informed by tools like SWIFT GPI (Swift, n.d.). If queued payment cancellations and expirations are rare, then participants can behave as if incoming queued payments were already in their account, a trust bolstered by Federal-Reserve-enforced stability of overnight secured lending markets like repo, allowing banks to continue queuing further payments, confident that they will be funded/netted out by queued incoming payments. After all, since The Clearing House does not take on liability, CHIPS would not be able to net payments so efficiently if participants did not significantly trust queued payments in this way, as netting depends heavily on the free flow of offsetting payments, which would not flow so easily if currently queued payments were unknown or unreliable. In this situation, there is no major economic difference between CCP net credits/debits and the queued payment net credit/debit construct, other than the markets they operate in. Thus queued net credit/debits satisfy the 3 key monetary properties of internal ledgers, by very similar logic as CCP net credits/debits, making them money for the interbank market.

1.2.10 On Simple Sum v.s Component Weighting

Up to this point I've implicitly implying there is a distinction between a definition of money and a monetary aggregate, but should there be? The divergence between a definition and an aggregate centers on weighting (and thresholding if the definition isn't precise enough). A monetary aggregate may include all the forms of money that pass the definitional test, but it may weigh them differently. In this section I will evaluate two weighting schemes, the first proposed by Barnett (1980) and the second by Spindt (1985a). I conclude that simple sum is actually preferable to both. The simple sum approach does not stabilize velocity, which will be addressed in a separate essay.

Divisia aggregates do not themselves determine the breadth of the monetary aggregate, since Barnett works with the micro-founded framework of money as a good which provides monetary services, like a housing good provides shelter services (which would not accept any obvious and discrete separation between an instrument like a Money Market Fund and a High Yield savings account, since they both provide similar monetary services to the individual). Consequently, the Divisia method can be applied to any level of standard monetary aggregation, and Barnett advocates for it's use at all levels above M1 (Barnett, 1980, pp. 13,19,35–36,39). Barnett defends

variable weighting by appealing to conventional index number theory, which when applied to money as standard good just like any other, claims that because different monetary goods are not perfect substitutes (due to liquidity differences and their corresponding compensating interest rates), weighting one dollar of a zero interest rate demand deposit the same as one dollar of a 5% savings deposit is economically unjustified (Barnett, 1980, pp. 14–15). He defines:

$$s_i = \frac{\pi_i m_i}{\sum_j \pi_j m_j}, \quad d \ln M^* = \sum_i s_i d \ln m_i \quad (\text{Barnett, 1980, pp. 38}).$$

Here,

M^* : the Divisia monetary quantity aggregate;

m_i : the real quantity (real balance) of monetary asset i (Barnett, 1980, pp. 16);

π_i : the user cost, or equivalent rental price, of monetary asset i , i.e. the price of the monetary services supplied by that asset (Barnett, 1980, pp. 19), with liquidity constituting an important source of those monetary services (Barnett, 1980, pp. 42);

s_i : asset i 's share of total user-cost-valued monetary services;

$d \ln m_i$: the infinitesimal proportional change in the quantity of asset i ;

$d \ln M^*$: the infinitesimal proportional change in the aggregate monetary quantity.

Essentially, Barnett is constructing a utility-equivalent quantity index for money. But is this the most macro-economically useful concept of a monetary quantity aggregate? It's certainly not the only coherent quantity aggregate/index that could be applied to a class of goods, although it surely is the most obvious one. For example, say we want to construct an aggregate of transportation vehicles. We want to understand how the quantity of transportation vehicles changes over time. We can't simply sum a diesel truck with an electric powered mega-yacht, so we need an objective way of weighing the relative quantities of each vehicle, based on their "importance". How do we determine that importance? The obvious way is by how much utility they provide to a representative consumer, interpreted through market prices. This would be a utility-equivalent quantity aggregate. But what if what we're studying the environmental impact of transportation vehicles? We want to be able

to distinguish changes in quantity of transportation vehicles from changes in environmental impact per vehicle. In this case, we shouldn't weight an electric mega-yacht more heavily than diesel truck simply because it costs more. Importance weighting of individual component vehicles would instead be based on environmental impact. Thus which is the correct quantity aggregate of transportation vehicles? The answer is that it depends on what question you are trying to address. If you want to best understand the relationship between quantity of transportation vehicles and gross nominal transportation vehicle expenditure, you want to construct a quantity-price decomposition and therefore a utility-equivalent quantity index, but there could be viable alternatives based on what the most important relationships involving that quantity aggregate are.

Therefore I propose the following logic for deciding the correct aggregator function given a vectorized concept c : among all the possible dual vectorized concepts d whose dot product with c has any meaning, the dot product whose relationship with c is the most important to understand determines the dual, factor-reversible decomposition which should define C (c 's aggregator). Furthermore, once the correct dual factor d is identified, the best decomposition is the one that best separates C from D . In other words, we want the decomposition of $\mathbf{c}^\top \mathbf{d}$ which most closely approaches an exact scalar factorization of the component-wise bilinear aggregate $\mathbf{c}^\top \mathbf{d}$, where C depends on c and D only depends on d .

So when c is m (the vector of money good quantities), what is the aggregate $\mathbf{m}^\top \mathbf{d}$ (and accompanying d) whose relationship with m is most important to understand? Naturally, that aggregate is something like nominal gross national product (PY from the equation of exchange). For this aggregate, the right d is v (the vector of money good velocities), according to the equation of exchange. This, however, is not the decomposition Barnett lands on: his chosen $\mathbf{m}^\top \mathbf{d}$ is a "total user-cost-evaluated expenditure allocated to passbook account services" where $d = \pi$ (the vector of user cost, or equivalent rental price of monetary goods) (Barnett, 1980, pp. 24,26). Should the relationship between money quantity and "total user-cost-evaluated expenditure allocated to passbook account services" really define the money quantity aggregator, or should the defining relationship rather be between money quantity and gross national product? Clearly the latter relationship seems to be more important, so optimizing the dual decomposition with regard to that aggregate should be the priority. Barnett himself defends Divisia aggregates due to increased clarity they provide in when analyzing the relationship between money quantity, velocity, and Gross National

Product, as opposed to simple sum (Barnett, 1980, pp. 39–41).

For Barnett, since he decides to conceptualize money quantity as a utility-equivalent quantity index, then it is indeed true that, as index number theory requires, component money good quantities can only be simply summed if they are perfectly substitutable, which they clearly are not since they have different attributes. However if we decide the relationship between money quantity and gross national product is more important than “total user-cost-evaluated expenditure allocated to passbook account services”, then we are not conceptualizing money quantity as a utility-equivalent quantity index, and thus are not bound by the perfect substitutability condition for simply summed quantity aggregates.

No matter the decomposition chosen however, as mentioned earlier, the ideal aggregator function would only depend on their own component vectors (and unknown parameters), as Barnett (1980, pp. 37) notes (but as Spindt (1985b) notes, this it is impossible to satisfy this condition as well as factor reversibility if the dual factors can be any positive vectors without structural constraints). A motivating reason is that this would ensure the cleanest separation between the two dual aggregators, so that changes in either summarize changes in their respective component vectors only, rather than being a pair of factor reversible aggregates which are just some arbitrary decomposition of the dual pair vectors dot product, each encoding too much information from its counterpart dual component vector. Practically though, Barnett must use index number approximations rather than the underlying functional aggregate, since index numbers depend on measurable prices rather than unknown parameters, and are also exact to the extent that the data is consistent with microeconomic maximizing behavior Barnett (1980, pp. 37–38). This means that Barnett’s utility-equivalent money quantity index encodes changes in π , the user cost, which of course, as just explained, is not an issue to the extent that economic data is consistent with microeconomic maximizing behavior, and to the extent that a utility-equivalent money quantity aggregator is the one with the properties we want. However in the following paragraph I aim to demonstrate that this π dependency/encoding does actually create issues, which show that the utility-equivalent money quantity aggregator does *not* have the properties we want:

While Barnett cautiously notes that “An economic reason for the heavy weighting of transaction balances is that their liquidity contributes heavily to monetary services” (Barnett, 1980, pp. 42), it shouldn’t be a stretch to say liquidity is by far the most dominant factor in what determines the value of a monetary service. Furthermore,

I don't think it's a stretch to say there should also be a very strong relationship between liquidity and velocity. After all, what's the point of liquidity if it's not being used? Yes, the relationship isn't necessarily one to one: it can be useful to hold a highly liquid money even if you don't take full advantage of that liquidity because the option itself is valuable. However, rational economic actors should optimize their sweeping behavior, whereby lower turnover funds should be held in lower liquidity money goods (and earn interest in compensation) to the point that we should expect the relationship between liquidity and velocity to be strong. Thus we can expect π and therefore s_i to encode a significant amount of information about money good i 's velocity. This means that a utility-equivalent money quantity aggregator is not a pure quantity aggregate relative to the quantity-velocity decomposition, and thus it mechanically hides destabilization in velocity as destabilization in the monetary aggregate, rather than truly improving the stability of velocity. In other words, while changes in this money quantity aggregate may have a more reliable relationship with GNP, that's not actually terribly useful, because the task of having to predict/explain instability in velocity of component goods did not go away, it's just part of predicting changes in the monetary velocity aggregate itself, rather than just impacting the velocity aggregate. Of course just because v may be heavily correlated with π does not change the fact that encoding the dual component vector π into the money quantity aggregate is justified for its use as a utility-equivalent quantity index (since prices can be used to approximate representative consumer utility which is used as objective importance weighting of component goods). However encoding v into M^* does potentially become questionable when using the aggregate to analyze money quantity in the context of relationships with velocity and gross national product, since the quantity-velocity decomposition is not the same as the quantity-price decomposition, and thus we no longer have the guarantee that encoding the dual component vector into the quantity aggregator function is a non issue, since objective money good weighting is no longer dependent on the representative consumer's utility function. This means that we cannot say changes in a utility-equivalent money quantity aggregator truly reflect changes in quantity as opposed to changes in velocity. Thus it runs the risk of no longer having a meaningful interpretation for the most relevant questions, potentially approaching an arbitrary scalar.

Let's now examine [Spindt \(1985a, pp. 178–181\)](#)'s proposed aggregators:

$$\frac{M(\mathbf{m}_1)}{M(\mathbf{m}_0)} = \left(\frac{\mathbf{m}_1^\top \mathbf{v}_0}{\mathbf{m}_0^\top \mathbf{v}_0} \cdot \frac{\mathbf{m}_1^\top \mathbf{v}_1}{\mathbf{m}_0^\top \mathbf{v}_1} \right)^{1/2},$$

$$\frac{V(\mathbf{v}_1)}{V(\mathbf{v}_0)} = \left(\frac{\mathbf{m}_0^\top \mathbf{v}_1}{\mathbf{m}_0^\top \mathbf{v}_0} \cdot \frac{\mathbf{m}_1^\top \mathbf{v}_1}{\mathbf{m}_1^\top \mathbf{v}_0} \right)^{1/2},$$

where $\mathbf{m}_t = (m_{1t}, \dots, m_{Kt})^\top$ is the vector of holdings (stocks) of the K means-of-payment goods in period t , $\mathbf{v}_t = (v_{1t}, \dots, v_{Kt})^\top$ is the corresponding vector of turnover rates, and the subscripts 0 and 1 designate the two time periods being compared. $M(\mathbf{m}_t)$ is the scalar money-stock aggregator and $V(\mathbf{v}_t)$ the scalar velocity aggregator, satisfying

$$M(\mathbf{m}_t)V(\mathbf{v}_t) = \mathbf{m}_t^\top \mathbf{v}_t.$$

Spindt takes the approach of aggregating money while looking through the more macro-economically useful lens of velocity rather than Barnett's user cost of money services. This seems to be the right approach: user cost of money services is like the lens of a magnifying glass, used by the individual economic actor to study their immediate local environment (their personal finances). At the macro level, however, properly surveying the economic landscape with a bird's eye view requires a different lens, the telescope of velocity. As Spindt points out 'In particular, the economic focus of the analysis underlying the Divisia aggregates is the demand for monetary services, and in this sense, a money good is analytically indistinguishable from any other capital good-autos, to choose randomly, which are demanded for "automotive" services' (Spindt, 1985a, pp. 177). There is a reason why money is one side of the equation of exchange while all other goods are on the other.

When introducing his alternative aggregators, Spindt (1985a, pp. 179) asserts 'In the particular solution derived in Spindt (1984), the Fisher ideal index number formula is exact for the aggregator functions $M(\mathbf{m})$ and $V(\mathbf{v})$ '. But the question again is, even though the index number result may be an exact approximation of the aggregators given cost optimization assumptions, are the underlying functional form aggregators actually describing money and velocity in the way that matters? Are they the best representations of those concepts, the best factor reversible pair with respect to a gross nominal spending measure, or are they just one possible pair?

In aforementioned letter, Spindt (1985b) models the economy as selecting an efficient trading sequence whose associated vector of money-good turnover rates is $\mathbf{v}$. Given the vector of money stocks $\mathbf{m}$ and a required nominal volume of trade z , where the equation of exchange requires $\mathbf{m}^\top \mathbf{v} = z$, efficiency is postulated to mean choosing

the feasible turnover vector that minimizes the real resources absorbed in trading:

$$\mathbf{v}^*(\mathbf{m}, z) = \arg \min_{\mathbf{v}} C(\mathbf{v}) \quad \text{s.t.} \quad \mathbf{m}^\top \mathbf{v} = z.$$

Here $C(\mathbf{v})$ is introduced directly as a primitive resource-cost schedule over turnover rates; it is not a conventional cost function. At the optimal solution, the corresponding first-order condition implies

$$\mathbf{m} \propto \nabla C(\mathbf{v}^*),$$

so the money-stock vector is proportional to the vector of marginal trading-resource costs associated with turnover. Spindt asserts the velocity aggregator equal to the cost schedule:

$$V(\mathbf{v}) \equiv C(\mathbf{v}),$$

and obtains the corresponding money aggregator residually so that, for efficient $(\mathbf{m}, \mathbf{v})$,

$$M(\mathbf{m})V(\mathbf{v}) = \mathbf{m}^\top \mathbf{v} = z.$$

Unlike standard index-number constructions whose underlying aggregators are given conventional utility, production, or production-cost interpretations, Spindt's $C(\mathbf{v})$ is not a conventional cost function and $M(\mathbf{m})$ is not derived as a conventional production function mapping productive inputs into maximal output.

The first issue that jumps out to me is the assertion $V(\mathbf{v}) \equiv C(\mathbf{v})$. It's a substantial logical jump without adequately explicit economic justification. Why should we think of aggregate velocity as aggregate trading cost induced by monetary exchange during a trading sequence? Maybe this would make sense if trading cost optimization were really what primarily drove velocity changes (given a target equilibrium gross nominal expenditure level z), but that points to the deeper issue: sure we can conceptualize a theoretical function describing the aggregate cost of monetary exchange, including things such as payment processor fees, which certainly depends on money velocity, but is the optimization process of minimizing those costs really the primary driver determining the distribution of velocity across different money goods and therefore "aggregate velocity", or is velocity more heavily determined by [likely to be mostly] independent factors such as consumer sentiment and risk appetite? It seems to me that equating velocity with these kind of "demand" concepts is far more defensible (since if economic agents in the entire economy are spending a higher percentage of their money at a rapid pace, then presumably their demand and/or

optimism and/or risk appetite has increased), and does not seem to be same thing as aggregate cost of monetary exchange.

There is another problematic result. Spindt's model assumes m does not change over the course of the trading sequence T (which cannot be instantaneous since it begins with an initial allocation of goods and ends in the equilibrium allocation of goods), which of course is completely unrealistic. This is not a harmless simplification, because half of the purpose of the model in the first place is to describe how money supply changes over time. Yes, Spindt can say there are gaps in the trading sequences he models and changes in m can be modeled as occurring in those gaps, but that is still not a sufficiently accurate model of the economy because money supply is constantly changing, there are no blocks of time where trading sequences occur yet money supply doesn't change. Either way, Spindt has no theory inside this model explaining how the money-stock vector changes between the observations to which his money quantity index is then applied. His model cannot be applied to periods of time during which m changes.

Obviously Spindt recognized this, and it's not immediately fatal, since the model could simply be one component of a more general equilibrium model where m is allowed to change wherein the trading cost optimization model still describes equilibrium v . However, in any general model that preserves the core logic justifying Spindt's aggregates, $\mathbf{m}_t \propto \nabla C(\mathbf{v}_t)$ must hold at every observation for the optimization argument rationalizing his aggregators to apply. But herein lies a glaring issue: bank deposits should be a significant portion of any reasonable monetary aggregate, and it's safe to assume that they should make up a significant majority of the one I will propose (which will contain all the instruments discussed in previous sections which pass the definitional test). Is it really true that the level of credit creation/destruction through loan origination/repayment is uniquely determined (conditional on v and z) by minimizing the costs of monetary exchange, costs such as payment processing fees, physical transaction costs, cash management costs, payments software, other operational costs involving payments including wages of employees in charge of payments, etc.? This seems far-fetched.

This result can be contrasted with ordinary consumer theory, in which the parallel first order condition is $\mathbf{p} \propto \nabla U(\mathbf{x})$, which means $\frac{MU_i}{MU_j} = \frac{p_i}{p_j}$, which has a clear and natural interpretation: At the optimum, the consumer's marginal willingness to substitute between goods equals their objective market exchange ratio. Spindt instead requires $\frac{m_i}{m_j} = \frac{\partial C / \partial v_i}{\partial C / \partial v_j}$ but why should the relative outstanding stocks of two different

monetary instruments equal the relative marginal real-resource costs of increasing their turnover rates, up to a common scalar?

Now consider the following factor-reversible, simple sum money-velocity decomposition:

Given

$$\mathbf{m}^T \mathbf{v} = \text{GNP} \quad (\text{equation of exchange}).$$

Let

$$s_i := \frac{m_i}{\sum_{j=1}^I m_j}.$$

Define

$$M := \sum_{i=1}^I m_i, \quad V := \sum_{i=1}^I s_i v_i.$$

Then

$$MV = \text{GNP}.$$

where

I := number of component money goods,

$\mathbf{m}$:= vector of component money-good quantities,

$\mathbf{v}$:= vector of component money-good velocities,

M := money quantity aggregator,

V := money velocity aggregator,

GNP := gross national product, or analogous gross nominal expenditure measure.

Note this decomposition has the following desirable properties:

- factor reversibility with respect to gross nominal expenditure (as opposed to something like Barnett's 'total user-cost-evaluated expenditure allocated to passbook account services')
- Monetary aggregate is not an index approximation of directly unmeasurable functions dependent on efficient optimization assumptions, but gives the actual directly interpretable level
- changes in M truly depend purely on changes in m ; it does not encode changes in the dual factor (which must be justified by efficient optimization assumptions).

- if using my precise definition of money, there is no additional work to be done from definition to aggregate, all instruments that pass the definitional test are simply summed.

Of course V still encodes m through the weighting of v by s , but this has a natural and simple justification: If agents are converting high velocity checking accounts into low velocity savings accounts, It would make sense to say that in aggregate, velocity in the system is decreasing, because there are less high-velocity accounts and more low-velocity accounts, even though the velocity of each individual account type does not change. The amount of dollars in the system hasn't changed, one dollar is still worth one dollar no matter if it's in checking or savings, what changed is their velocity attribute, captured by the velocity aggregate. This gets to the intuitive justification of equally weighting different money goods that no doubt have nonidentical attributes: dollars are contractually pegged to be convertible one for one; different forms of dollars have fixed exchange rates/relative prices (exceptions include early withdrawals from time deposits, but they can still be weighted equally because we can make a small assumption that in the vast majority of cases, agents plan well and do not withdraw early. Negotiable CDs would need to be weighted by their below par market prices). Converting from one type of money to another does require a market transaction (given my definition of money and other than the negotiable CD special case), unlike any other conversion between a money good and a non monetary good, or two non monetary goods. Interest rates and liquidity restrictions are intrinsic properties of money, not money's price. It's not necessary to model those properties independently of money supply and velocity: they only matter, from the perspective of the equation of exchange, to the extent that they affect money supply and velocity. We are justified in simply summing the various money goods even if they don't satisfy they aren't perfectly substitutable, since the most useful concept of money quantity is not a micro-focused utility-equivalent quantity concept, but a macro-focused aggregator defined by the quantity-velocity decomposition. Rather than being perfectly substitutable, distinct money goods instead satisfy a closely related criterion, "equilibrium marginal equivalence":

$$\text{MRS}_{ij}(m_{it}, m_{jt}; \theta_t) = 1 \quad \forall i, j, \forall t,$$

where:

$m_{it} \equiv$ the quantity of money good i held at time t ,

$m_{jt} \equiv$ the quantity of money good j held at time t ,

$\theta_t \equiv$ the vector of intrinsic liquidity restrictions and interest-rate properties of the money goods at time t .

Now I'm going to sidetrack briefly to defend the following claim, partially motivated by the need to sharpen the defense of my quantity-velocity decomposition: *velocity is equivalent to real demand* (Under this interpretation, the equation of exchange is a decomposition of the accounting identity *nominal demand = nominal supply (nominal value of real supply)*). The intuitive reasoning for why velocity represents real demand is simple: If velocity increases, agents are using their money more actively to make purchases; they're exercising a higher portion of their latent purchasing power, and vice versa. If that's not demand, I don't know what is. Conventional macroeconomic theory defines real demand as *quantity of goods/real output desired*, which must always equal quantity of goods supplied, in equilibrium. The theoretical aggregate demand schedule can describe the broader structure of demand at different price levels away from what is actually observed, but the output is still *quantity of goods/real output desired*. This concept of real demand is useful to model how demand initiated movements can push prices from the old equilibrium to the new, but real demand cannot move independently from real supply, outside of mostly minor divergences during disequilibrium (if we can assume that generally speaking, markets are efficient and thus most of the time, at any given time, prices are very near equilibrium).

The fact is however macroeconomists like to separate demand side effects from supply side effects, demand shocks from supply shocks, as if they were two completely independent and fundamentally different variables. That stark differentiation is definitely somewhat justified by the conventional conception of real demand, since it supports the identification of the causal initiator of a new equilibrium, but since when real demand and real supply are plotted over time it's essentially the same line (to the degree that markets are in equilibrium), something is left to be desired. For example, we should be able to say: "real equilibrium demand from t_0 to t_1 fell to a lower level, and so prices fell to maintain equilibrium, but real supply remained unchanged". I propose velocity as an additional conceptualization of real demand, not as a redefinition of the current conceptualization as *quantity of goods/real output*

desired, but as a new concept that can move independently from real supply, even in equilibrium: *real demand for nominal supply (PY)*. This doesn't break the equilibrium *demand = supply* condition, because *nominal demand = nominal supply* still holds: $MV = PY$. Neither does it change the fact that *real demand for real supply = real supply* at equilibrium.

This version of real demand as velocity would naturally be measured as a rate: turnovers per period. General turnover isn't equivalent to the proper concept of velocity for the equation of exchange, as general turnover includes non-final transactions, which can inflate real demand just due to an increase in intermediate transactions passing the same newly produced good around, and also includes second hand transactions. Thus some empirical estimate of turnover for final sales would need to be derived, or would just be residually defined from the equation of exchange, depending on the situation. The full scope of the implications of velocity as real demand are beyond the scope of this paper, but it helps to illuminate why my quantity-velocity decomposition is structured the way it is:

Changes in supply are typically more constrained than changes in demand. Agents can simply decide, on a dime, to stop discretionary spending, whereas a factory cannot simply wish more output into existence. The simple sum monetary aggregate reflects this "sticky" property of supply: it doesn't change simply because economic agents wake up one day and decide they want to convert 90% of their demand deposits into savings. If this happens yet v remains unchanged; the shift from demand to savings deposits should still be modeled as a decrease in velocity, i.e real demand (for nominal supply), since it can happen on the whim of deposit account holders. Thus the encoding of changes in m into V through weights s is perfectly natural and requires no optimization structure to justify. Changes in aggregate money supply should only reflect changes in more "sticky" factors, such as deposit creation/destruction through loan origination/repayment. This is exactly how a simply summed money aggregate behaves.

2 Inflation

There's a divergence between formal definitions of inflation (any increase in price level), its rhetorical/implied meaning in market commentary (broad, sustained, tied to monetary reasons), and indices measuring it. The goal of [section 2](#) is to propose a definition that can align all three.

CPI-U is a price index designed to estimate the minimum cost of maintaining a constant utility level for the average urban consumer (U.S. Bureau of Labor Statistics, 2018, pp. 2–3). It’s the most widely cited figure used to describe broad, general, economy wide inflation. While many professionals are aware that other indices are often more appropriate depending on context, CPI-U is still often used because it’s the most well known. However, even these alternative deflators are not designed to capture inflation in investment assets, which are a key component of the opportunity set of the dollar, and are thus necessary for a true general purpose inflation measure. Even CPI-U is handicapped for its own stated purpose (describing price inflation for goods/services the typical urban consumer needs to maintain a constant utility level) by not including investment asset inflation, which crucially includes real estate. Real estate price inflation is only indirectly included as rent inflation through imputed rent. Bolhuis et al. (2024, fig. 7) demonstrates that the pre 1983 CPI-U method, which does not use imputed rent, plotted from 1983 - 2023, is typically higher than official CPI-U in that same period. The ability to invest is an important component of the average consumer’s utility function, particularly owning a home, yet it’s not tracked directly by CPI-U.

One could argue that measuring purchasing power for investment assets is actually irrelevant, since the only purpose in purchasing them is to increase purchasing power for real goods, and purchasing power for real goods is tracked in CPI-U and others. Yet this is an oversimplification which misses a key temporal aspect: the utility of investment assets comes not from the ability to increase purchasing power for real assets *today*, which is tracked by conventional measures (and cannot be achieved through investment anyways), but the ability to increase purchasing power for real assets *in the future*, i.e. the ability to defer real consumption into to the future and be rewarded for it, whether for yourself or others. Thus focusing only on purchasing power for real assets will ignore purchasing power for a major component of consumer’s utility, the utility of investment. Changes in real goods purchasing power in the future do not capture changes in ability to effectively invest today. One cannot rely on purchasing power for real goods at any time t to reflect purchasing power for all utility at t , because if, for example, purchasing power for investment assets at t_0 was low, then the ability to fully reap the benefits of increased purchasing power for real assets at time t is diminished, since deferring consumption to the future was unattractive at t_0 , resulting in fewer funds to consume better priced real goods at t , resulting in a decrease of utility which is not captured by a real goods-only inflation

measure. A second reason to exclude speculative investment assets from inflation, is the fact that their prices are volatile, speculative, based on future expectations, etc., which make price movements hard to interpret and pin down on any one cause, whereas consumer goods prices are more interpretable, and subject to simpler supply and demand dynamics. While this is true, the alternative is complete blindness to investment utility, a major component of total utility. If money begins to flow into investment assets rather than real goods at an unusually rapid pace, then a general inflation measure without investment assets would incorrectly say the price level has gone down and purchasing power has increased, when in reality demand has just shifted from real to investment assets. A third reason for not considering investment assets in inflation, specifically financial assets, is that every financial asset is someone else's liability and some represent ownership over real goods, and thus considering them is just double counting, since financial assets and liabilities net out to zero if the entire world is considered. However this is not a valid reason to exclude financial assets. If I take out a new loan and the bank securitizes that loan and sells it, just because the security owner's asset cancels out with my liability does not mean a new asset has not just been created. Before I took out the loan, there were one fewer debt contracts on the market. The loan holder did not have the opportunity to buy the rights to my future payments before the security was created. Excluding financial assets like these from inflation ignores a meaningful chunk of the dollar's opportunity set that is not represented by other goods. Even if the loan is collateralized, the claim to rights to collect future payments is an asset distinct from the collateral. The fact that a new liability has been created does not change the fact that a new asset has been created as well, and that asset's price must be considered in a valid general inflation measure. Equity is more complex because a portion of its value is just a reshuffling of ownership of existing real goods, but the other portion is a claim over rights to future earnings which is not represented by other assets. Thus not only does CPI-U fail as a general purpose inflation measure, but so do any other commonly used price indices such as those for GDP or Gross Domestic Purchases.

Ideally, the definition/metric of inflation should line up with the rhetorical way in which it often get used, which is a broad, sustained decrease in purchasing power across most if not all goods, originating from a similar cause, not simply any transitory change in price level due to some idiosyncratic feature of a particular good or market. The definition I propose is designed to capture the force described in rhetorical use, not just any change in price level. It's based on the premise that at the

secular aggregate level, supply of money, supply of non-monetary goods, and quality of non monetary goods, are the only factors significantly influencing prices, implying major aggregate changes on the demand side are only temporary and cyclical, staying bounded within a relatively tight range. In other words, I assume *stable real demand (for nominal supply)* - which may be empirically questionable within contemporary economies distorted by central banking. The mechanism which should stabilize real demand, and why central banking destabilizes it, is outside the scope of this essay.

2.1 Unit Elasticity of Diverse Aggregates

Beyond *stable real money demand*, a second (and closely related) testable requirement for my definition of inflation is “Unit Elasticity for sufficiently Diverse Aggregates” (UEDA), meaning: the broader/more diverse product aggregate is, the closer its Share-Weighted Mean of Individual Realized Price Elasticities (SWMIRPE) is to unit elasticity. The intuitive justification is simple: *a priori*, extremely elastic or extremely inelastic demand are both highly unlikely, and “normal” unit elasticity is most probable for any given product. Yes, there are complex reflexive interactions and feedback loops wherein a product price depends on other prices, not just on its own supply and elasticity to that supply, but for a large and diverse aggregate those forces may cancel each other out.

I could find no theoretical support or empirical evidence that directly addresses this question, so I attempted to empirically verify it myself, with heavy reliance on artificial intelligence, as I am very much out of my depth here. I prompted *GPT-5.4 Extended Thinking* to locate publicly available data required to test this UEDA empirically, and to find existing academic work describing how to construct elasticity models that could be used to calculate SWMIRPE for the broadest product aggregates possible. I then prompted it to construct its own prompts which were passed on to *GPT-5.3-Codex*, which conducted analysis using python. The results were fed back to *GPT-5.4 extended thinking*, which interpreted them, finding issues and offering improvements which were passed back to *GPT-5.3-Codex*. The final results as summarized by *GPT-5.4 Extended Thinking* are:

“Using a paper-specific hierarchy that combines Shojaeddini et al.’s top-tier full-consumption demand systems (QUAIDS, AIDS, LES), a Jorgenson-Slesnick-style stage-2 rank-2/rank-3 exact-aggregation benchmark, Döpper-inspired modular heterogeneity diagnostics, and an Inforum/IdLIFT/Chao reduced bridge-matrix plus input-output propagation framework, the most credible conclusion of the exercise

is that the share-weighted mean of individual realized own-price elasticities across broad U.S. market consumption categories is very near unit elasticity, with the preferred headline estimate equal to -0.9544 for the 8-category market-goods benchmark; the corresponding preferred high-plausibility Marshallian weighted mean is -0.9420. This conclusion is intentionally narrower than a full-consumption statement “across all goods including leisure,” because the leisure block-constructed as imputed expenditure equal to leisure hours times an instrumented wage and priced by a wage proxy-proved to be the main source of instability: in the leisure-inclusive top-tier QUAIDS specification, leisure carried a dominant expenditure share of about 0.748, had a positive own-price elasticity of about 2.011, and on its own contributed about +1.505 to the weighted mean, which is why the leisure-inclusive aggregate became implausibly positive and had to be downgraded. After excluding or neutralizing leisure and focusing on high-plausibility market-goods subsets-Subset A excluding leisure entirely, Subset B requiring sign stability, and Subset C requiring both sign stability and stronger price/bridge quality-the realized estimates clustered tightly around unity in magnitude (roughly -0.95 to -0.97), with the final benchmark dominated by economically interpretable categories such as housing, food and beverages, medical care, and transportation rather than by fragile imputed components. Accordingly, the bottom-line empirical thesis is not that there is a theorem forcing economy-wide demand toward unit elasticity, but that, for broad market-goods aggregates measured with plausible shares, prices, and IO mappings, the best current evidence from this pipeline places the share-weighted mean of realized own-price elasticities close to -1, while full-consumption estimates that include imputed leisure remain too specification-sensitive to serve as a reliable headline fact.” Code Repo: (Kofman, 2026b), Final Report: (Kofman, 2026a), elasticity model theory: (Shojaeddini et al., 2021; Chao, 1991; Jorgenson and Slesnick, 2008; Inforum, n.d.a,n; Döpfer et al., 2024; Meade, 2001b,a)

2.2 General Inflation Definition

Subsection 2.1 is intended to provide defense to support potential plausibility of UEDA, not to prove it. Taking the empirical validity of these properties as given, a general inflation definition can be formalized. Step one is to simplify *stable real demand (for nominal supply)* by initializing a constant “demand” variable, since we mostly care about persistent secular effects on price level, not perfect accuracy at every point in time. D is essentially a reformulation of money velocity, defined by

the ratio of “terminal sales” (nominal total last sales in a given period of all goods, including investment goods and second hand goods) to money supply, rather than final sales (used in a measure like gross national product) which do not include investment goods or second hand goods. The reason for including second hand goods is that some, like real estate, may not grow very much yet are critical to total utility, and thus are under weighted in measures such as gross national product.

τ := fixed window length

$\mathcal{T}_\tau(t)$:= terminal sales over $[t - \tau, t]$

$M(t)$:= money supply at time t

$$\text{Change ratio between } t_0 \text{ and } t_1 = \frac{\mathcal{T}_\tau(t_1) - \mathcal{T}_\tau(t_0)}{M(t_1) - M(t_0)}$$

$$\text{Point-in-time ratio at } t_0 = \left. \frac{d\mathcal{T}_\tau}{dM} \right|_{t=t_0} = \frac{\dot{\mathcal{T}}_\tau(t_0)}{\dot{M}(t_0)}$$

$$\frac{d\mathcal{T}_\tau}{dM} = D \quad \text{for all } t \tag{1}$$

D := constant ratio of changes in windowed terminal sales to changes in money supply

Given D , inflation can be derived from the following identify:

$$M_t D = Q_t Z_t \Psi_t \tag{2}$$

D : As defined previously. τ can be arbitrarily set to make the relationship hold. The stable demand assumption asserts it can remain near constant without violating the equality.

M_t : money supply at time t

Q_t : stock index of all goods quantity at time t

Z_t : utility per good index at time t

Ψ_t : price per unit utility index at time t

“Unit utility” is designed as a *common, objective, intrinsic-value unit construct* that each good can be converted into / represented by in order to understand trends in purchasing power across all goods, weighted by importance.

UEDE is implied by 2, when money supply is held constant to isolate the effects of quantity changes on price:

$$C = Q_t P_t \tag{3}$$

$$\ln C = \ln(Q_t P_t)$$

$$= \ln Q_t + \ln P_t$$

$$d(\ln C) = d \ln Q_t + d \ln P_t$$

$$0 = d \ln Q_t + d \ln P_t$$

$$d \ln P_t = - d \ln Q_t$$

$$e := \frac{d \ln P_t}{d \ln Q_t} = -1$$

e : SWMIRPE

C : MD

P_t : $Z_t \Psi_t$, average price per good

In addition to the variable M form of 3, 2 can be expressed as:

$$M_t D = S_t \Psi_t \tag{4}$$

M_t : money supply at time t

S_t : quality-adjusted ”supply” index at time t

Ψ_t : price per unit utility index at time t

From here inflation can be defined as the change in Ψ_t . If inflation is defined as simply the change in general price level, then P_t would be the correct object, but what actually matters is purchasing power for economic utility, not units of goods.

Those may be the same object in many situations, but just focusing on price per unit good ignores improvements in quality, which is why official quantity indices adjust for quality.

$$\begin{aligned}
M_t D &= S_t \Psi_t \\
M_{t-1} D &= S_{t-1} \Psi_{t-1} \\
\frac{M_t D}{M_{t-1} D} &= \frac{S_t \Psi_t}{S_{t-1} \Psi_{t-1}} \\
\frac{M_t}{M_{t-1}} &= \frac{S_t}{S_{t-1}} \cdot \frac{\Psi_t}{\Psi_{t-1}} \\
1 + m_t &= (1 + s_t)(1 + \psi_t) \\
1 + \psi_t &= \frac{1 + m_t}{1 + s_t} \\
\psi_t &= \frac{1 + m_t}{1 + s_t} - 1
\end{aligned} \tag{5}$$

m_t : rate of change in money supply,

s_t : rate of change in quality-adjusted supply,

ψ_t : rate of change in price per unit utility

2.3 General Inflation Metrics

Aggregating money supply was addressed in previous sections. With regard to s_t , for many goods, directly measuring quantities is unfeasible, and un-alike goods (adding apple quantities with car quantities) have to be coherently combined into a single quantity change rate, along with quality adjustments. This is addressed by federal statistical agencies for some non-speculative real good aggregates with quantity indices, which employ highly complex models, often combining nominal spending with individual goods prices to separate s_t from ψ_t ([Bureau of Economic Analysis, 2024](#), p. 4-16). Changes in q_t for the remaining major investment asset sectors: land, equity, and debt, are reasonably well defined concepts that can be measured directly, but the fact that investment asset quality is based on unknown, capitalized, expectations of future cash flows and resale price, measuring true z_t very difficult (to be clear, z_t is

the rate of change in investment assets utility/quality per unit good, meaning change in expected risk adjusted present value returns themselves, not the ratio of those returns to the good's price). Theoretically, if z_t estimates can be properly determined for large investment asset aggregates, those sectors can be combined coherently along with the existing non-speculative real good aggregates and their quantity indices, into a single average price level change rate, weighting sector quantity change rates by gross nominal value of domestic, nonbank holdings of these goods, and by a comparable synthetic stock-analogous value measure for flow sectors such as nondurables and services. While stable aggregate expenditure demand may not hold for each sector individually, especially investment assets, it doesn't matter if s_t can be properly measured for each and combined into an *all goods* s_t metric, assuming stable aggregate expenditure demand holds for the broadest *all goods* metric.

Including investment asset sectors into the *all goods* metric does potentially destabilize the stable aggregate expenditure demand, since investment asset prices are determined by unknown, capitalized, expectations of future cash flows / resale price: V_t (a similar concept to Z_t where Q_t is incorporated into expected resale price). This means changes in cashflows or q_t have no effect on ψ_t , if they were already expected at $t - 1$, which could invalidate the $\psi_t = \frac{1+m_t}{1+s_t} - 1$ relationship for the broadest *all goods* aggregate, since Fisher-style present value / intertemporal equilibrium setups imply this relationship does not apply to investment assets, which are a significant component of *all goods*. Not only do these models imply that expected/unexpected changes in Q_t do not directly impact the investment asset's valuation, but even when V_t changes (the expected future value based on expected future cash flows and resale price) this does not effect ψ_t either. The reason that q_t , expected or not, and v_t (rate of change in V_t), do not affect ψ_t is because D is not actually a constant but a changing variable for investment assets. In other words, holding M_t constant, if quantity of the investment asset increases, as long as its expected future cash flows and resale value are the same, price is unaffected because demand increases to absorb new issuance. Similarly, again holding M_t constant, if for example future cashflows are expected to be lower than they were previously estimated, then demand decreases and the price of the good will fall, but price per utility will remain at 1. After all, if price per utility increased, which would have to happen under $\psi_t = \frac{1+m_t}{1+s_t} - 1$, then people would realize the good is overpriced and there would be an arbitrage opportunity to short it, and thus the market would not be in equilibrium, and $\psi_t = \frac{1+m_t}{1+s_t} - 1$ would be inaccurate. If D is not constant for investment assets, it may not be constant for the

all goods aggregate which contains investment assets.

However D for the *all goods* aggregate can still be stable for a number of structural reasons (beyond the typical imperfect-market frictions that could stabilize D for investment assets): 1) Changes in D for investment assets in response to expected quality changes can come from complementary decreases in D for other sectors in a way that keeps overall D for *all goods* stable. Essentially, a significant portion of arbitrage capacity can come from changing consumption patterns from other goods, rather than tapping into or depositing into a designated dry powder reserves of money sitting ready to take advantage of investment opportunities (MMFs and treasuries are other investment assets, not money). 2) While demand is likely to be variable for smaller sub-components of investment sectors as they become more substitutable, the larger the aggregate becomes, an emergent phenomenon can occur as substitutability reduces. For example, while fixed income investors may be very willing to trade a bond from one company with a bond from another company in response to changing *risk adjusted present value returns* between them, they may be less willing to substitute their demand for the entire fixed income market with low-yielding money (especially considering FDIC insurance is more difficult to take advantage of for large sums). They may be more willing to adjust their risk appetite, in response to changing future cashflow expectations of assets, rather than change their demand for entire investment sectors, even as quality changes significantly for those sectors. Humans are creatures of habit, and it seems plausible there may be a significant habit for investing which is more steady than risk appetite, especially when everyone else's risk appetite changes in sync with yours, and it's difficult to even realize that it has changed, especially when you're convinced the market knows best and would arbitrage away any mispricing. This is classic bubble behavior which has been put on display many times, from the roaring 20's, Nifty Fifty, S&L crisis, dotcom, GFC, AI / Private Credit / Private Equity bubbles . . . Beyond individual psychology, many large institutions have mandates to achieve a certain yield, and their tendency to "reach for yield" when opportunities are scarce (Barbu et al., 2021) demonstrates the flexibility of risk appetite, rather than investment demand. 3) While risk appetite may not explicitly change, it may be subtly justified by adjusted modeling assumptions. Future expectations themselves are inherently unstable because the future is hard to predict, even with sophisticated mathematical models. Investing comes with inherent uncertainty, and for this reason future expectations may be influenced by the need to justify investing habits / mandates, and by the restrictions of binding balance sheet

/ trading constraints.

Thus one should not immediately assume D is unstable even if it includes investment assets. If it's stable, then we can proceed with identifying s_t for investment assets, so that it can be integrated into s_t for *all goods*. s_t is easier to conceptualize by separating into z_t and q_t . As mentioned earlier, q_t is more directly measurable, while measuring z_t (rate of change in utility/quality per unit good, i.e. change in expected risk adjusted present value returns) properly would involve sophisticated financial modeling to determine quality averages for large investment asset sectors, which could be informed by market prices to some degree but should not wholly rely on them due to potentially significant limitations in the flexibility of D , as described in the the previous paragraph. Subsection 2.3.7 is dedicated to starting points on measuring z_t for investment assets, but an actual complete model is outside the scope of this essay. s_t will be addressed for each investment sector in their respective sections.

These investment asset sectors can then be combined with other top-level sectors, by determining relative importance weights, based on the spending of nonbanks (in order to be consistent with the money supply measure), and should be isolated to domestic holders. The reason for the latter is: a currency's inflation measure's purpose is to describe purchasing power of income earners of that domestic currency, not to be an abstract concept of purchasing power of the currency itself with respect to strictly goods that can be directly purchased with that currency, which is less meaningful in a world with FX markets. The former assertion implies that spending of domestic currency income earners on foreign goods should be included as a sector, while spending of foreign currency income earners should be excluded from all sectors.

While the highly volatile nature of investment assets can distort their relative importance in the supply change rate, this can be smoothed out by using long run historical averages. After identifying the correct sectors and measures of their relative importance based on gross market value or sufficiently similar measures, their weights can be determined using the Fisher index method of combining Laspeyres (beginning of period market share) and Paasche (end of period market share) weights. As a sidenote, it's not inconsistent to refrain from depending on P_t to determine $z_t/q_t/s_t$ for lowest level components, while simultaneously using gross market value to determine importance weights for aggregation of components. If Z_t for a component decreases while P_t increases, it is fully consistent to say the impact/importance on the economy of movements in that component's Z_t are becoming more pronounced even if its S_t is

decreasing.

A practical highest level sector breakdown is: Consumer Durables, Investment Durables, Inventories, Consumer Nondurables, Consumer Services, Land, Share of Franchise Value, Debt, Foreign investment assets, Collectibles/Commodities/Art/Crypto, and Government Consumption/Investment. The following sections describe how to determine sector weights and measure s_t for each sector.

2.3.1 Consumer Durables, Investment Durables, Inventories

Each of these sectors have official federally published stock series and quality-adjusted quantity indices. An important caveat is that consumer and investment durable stocks are measured using current replacement cost bases, rather than direct market value (U.S. Department of Commerce, Bureau of Economic Analysis, 2003, pp. M-6, M-8), which can be systematically lower than market value because it doesn't account for replacement frictions like time to build and other intangibles, which are hard to estimate (Hall, 2001, p. 1185).

2.3.2 Consumer Nondurables and Non-OER Services

Consumer Nondurables and Services are consumed rather than accumulating as a stock. Thus determining their relative importance weighting (in terms of nominal spending) to the other sectors, which are measured as stocks, cannot be done directly. Construction of a synthetic importance weighting which is compatible with stock measures is required. This can be done by utilizing CPI-U, which has determined the correct relative flow weighting between Nondurables/Services and Durable goods, which does have a stock measure. By identifying the ratio of Nondurables and Services CPI-U weights to the durables CPI-U weight, and multiplying this ratio by the *current replacement cost basis* stock value measure of existing durable goods, it's possible to translate flow-based importance ratios into coarse synthetic stock levels by multiplying with the stock level of the durables component.

Both Consumer Nondurables and Services have existing quality-adjusted quantity indices, however CPI-U includes Owner's Equivalent Rent (OER) into services post 1983, while pre-1983 it employs a mixture of categories that attempt to model the same concept, but with more dependence on prices directly (Ptacek and Rippy, 2013, pp. 1–2). OER captures how much homeowners pay, or would pay if they weren't a homeowner, for the ability to live in their house (Ptacek and Rippy, 2013, pp. 1–2). This concept (and pre 1983 OER-like components) must be removed from both the

official quality-adjusted quantity indices, as well as from CPI-U (when calculating the flow-based importance ratio), because this expenditure is already tracked in *Investment Durables*, through housing structures prices, and by *Land* (otherwise it would be double counting).

2.3.3 Land

At the highest level of abstraction, there are two types of land: public and private. Public land can be treated as effectively off the market, and can contribute to overall private land q_t through net square footage entering/leaving the private market. Private land can further be broken down into undeveloped land, agricultural land, and developed land. Developed land can be further broken down into different zoning types, geographical areas, etc. q_t changes for each of these component goods can be measured again as net square footage entering/leaving the component, with each component's q_t can be weighted by its relative gross market value. Gross market value can be estimated using available national accounting, by summing the value of all real estate, residential and commercial, and subtracting the value of all structures. Land z_t does not really change. In national accounting, the value of land does not include crops or structures built on top of it, just the value of the plot itself (OECD, 2025, p. 33). The quality of the physical land itself does not fundamentally differ significantly from location to location, and neither does the quality of land development. Characteristics of land that most strongly affect its value, such as proximity to active economic zones, zoning/permitted density, infrastructure access, utility hookups, flood/fire risk mitigation, site preparation/topography, environmental constraints, etc. can largely all be captured by division into separate component goods whose weighting is determined by market value of the land within that component good.

For this reason, land should not be treated as a real good, but rather an *investment* good, meaning its utility is defined by its expected future cash flows and resale value. Any “real” value of the land is captured through the structures/crops built on top of it. Edge cases include privately owned (public land is not on the market and thus irrelevant to inflation) wild land used for recreation like hunting/fishing, in which case a portion of this land’s value is not dependant on its investment utility, i.e. expected future cashflow / resale value, but rather it’s “real” utility.

2.3.4 Share of Franchise Value

The component of equity value derived from *Share of Current Net Asset Value* (SCN) represents a claim on goods already tracked by other sectors (Investment Durables, Inventory, part of Land, part of Debt) and should not be double counted. The component of equity valuation not tracked by other sectors is the speculative share of expected future profits/NAV (Net Asset Value) valuation i.e. *Share of Franchise Value* (SFV).

Negative IMA Net Worth ([Federal Reserve Bank of St. Louis, 2026y](#)) is Franchise Value. If [negative IMA Net Worth] is negative, it means Franchise Value = 0 (unless the firm's non-equity liabilities exceed goods, in which case all outstanding equity value = SFV, because the entire value of the firm is based on expected future cash flows rather than existing goods. However this has never happened for the nonfinancial sector as a whole ([Federal Reserve Bank of St. Louis, 2026b,q](#))). In this situation, the underlying goods tracked in other sectors do not change from their recorded national accounting value. For publicly traded firms, the obvious SFV quantity measure is shares outstanding. However, for any one company, even though z_t can change based on earnings forecasts, the quantity of SFV always adds up to 100%, no matter how many shares are issued or bought back. What happens by the direct issuance of new shares is that z_t of existing shares is reduced by the same amount that q_t of shares increases, so s_t , ultimately the relevant object for inflation, is unchanged. Share issuance can be used to fund new projects which changes profitability, but this is captured by changes z_t , not q_t . A genuine measure of q_t for publicly traded firms is therefore net listings, which can change independently of z_t . Each firm could be treated as a unique good and weighted by its market cap, or aggregated by industry.

For privately held equity, the same SFV concept exists: the value of ownership stake in the expected future profits/NAV of the company beyond current NAV. q_t for private firms is analogous to net listings: Net firm entry (firm creation - dissolution), again weighted by estimated SFV. Even though these firms are not publicly traded, their SFV is still privately bought and sold at an economically significant scale. Valuation data is very opaque or nonexistent, but should be estimated (at the level of aggregate industry/sector averages) based on available information. An exact method for this is beyond the scope of this essay.

2.3.5 Debt

Debt is a unique asset category that has many properties making it money-like. However, fundamentally debt is not money, because it's not a medium of an exchange, as explained in [section 1](#). Debt must be purchased with money just like any other good. Thus, the same basic supply forces apply as with other goods, to the extent that D for debt is stable. If the supply of money increases while the supply of borrowers willing to take on debt stays the same, the price of debt will increase, generally meaning lower yields. On the flip-side, if money supply stays flat and the supply of debt increases, debt prices will go down. The unique/reverse devaluation impacts of inflation on debt's basic supply-price dynamics are best conceptualized as a component of z_t , not q_t .

A starting point for measuring debt q_t is raw nominal net principal changes (issuance-retirements/paydowns/redemptions), but it's too crude to simply aggregate tomorrow money and year 2100 money. It doesn't make sense to deal with this in the generic way other unlike goods quantities are dealt with (like apple vs car quantities or shares from different companies), which is to segment/weight quantity change rates by gross market value of the good, because the differences between dollars at different points in time are continuous, and they can be seamlessly combined into one quantity using present discounted value formulas (assuming other risk variables are same/similar). Gross duration-weighted nominal net issuance is also a better quantity measure than simply contract count, because what really matters is the size of those contracts, not their count, and it does not make sense to group contracts of different sizes into separate discrete categories, since those size differences are continuous. Thus Nominal net duration weighted issuance is the cleanest q_t measure, but segmentation is still required for different categories of debt. A potential top level split within the debt category could be: Treasury Securities, Agency/GSE backed Securities, Municipal Securities, Corporate and Foreign bills, Open market paper, Repo, and "Other loans and advances", as these are all mutually exclusive categories under national accounting ([Board of Governors of the Federal Reserve System, 2026b](#), pp. 31–34). Loans originated and held on bank balance sheets that are not available for trading in the nonbank market, should be excluded, which includes a (directly) undetectable portion of the "Other loans and advances" category.

2.3.6 Other

Crypto, Collectibles, Commodities, Art are relatively small sectors of the opportunity set of money, as measured by gross market value, are difficult to track. The effects of money supply become complicated when they're denominated in a wide range of different currencies with different supply growth rates. It's safe to ignore them (for now).

Government consumables and durables have their own FRED supply indices and can be included or depending on intended goals. While not exactly part of the money's opportunity set for the private sector in the direct sense, government expenditures are, in theory, essentially forced purchases that benefit the private sector and are merely organized by the government as the middle man.

Foreign investment assets are a significant part of the domestic currency's opportunity set and ideally changes in their supply should be considered as relevant factors to the domestic currency's purchasing power, barring concerns of complexity and limited data. These assets should fall under one of the previously mentioned categories and their supply measured accordingly. A caveat for these assets is that although, generally speaking, lagging price effects should average out over long term horizons, for which my inflation definition is designed, if consistently and explicitly taken advantage of by foreign central banks (China) through manipulation of currency supplies, then domestic inflation may be impacted in a way not tracked by my inflation definition, since it does not take into account foreign currency supply.

2.3.7 Financial Asset Quality Adjustment

The utility of SFV, debt, and land, Z_t , is defined by their expected risk adjusted present value returns. As explained in [subsection 2.3](#), asset quality can diverge substantially from P_t due to changes in Ψ_t , to the extent that D for these goods is unresponsive to z_t . Thus, under this inflation framework, P_t cannot be directly relied upon to determine Z_t . Instead, robust fundamental/credit analysis models designed for large aggregate sectors are required, but that's outside the scope of this essay.

An alternative approach to separating the empirical object q_t from the theoretical quality/risk forecast z_t (it must be a forward looking forecast rather than a simple backwards looking sharpe ratio so that it does not move erratically with market crashes, modeling bubbles as increases in investment Ψ_t which decreases below 1 and reverts back to 1 during the crash and subsequent recovery), is to consider the

combined object s_t . This framework would depend on forecasting risk adjusted cash flows for large investment sectors and modeling s_t as differences between the modeled cash flows for the existing goods in a given sector at the beginning of a period, and the realized cash flows for that good at the end of the period, including cash flows from the period's net newly created goods. In addition, s_t would also have to depend on changes in the beyond-period cashflow forecast of the assets comprising the sector at the beginning of the period, with the updated forecast of those initial assets, combined with the beyond-period cashflow forecast of any net newly created goods within the period. These two differences must be combined into a single s_t figure by weighting each appropriately, which depends on how the valuation date is set. If the date is naively set to one single time within the period, then s_t will be horizon dependent. The solution is:

$$1 + \dot{s}_{b,t}^{(\Delta)} := \frac{\mathcal{C}_{b,t,t+\Delta}^{\text{real}} + \mathcal{F}_{b,t+\Delta}^{\text{new}}}{\mathcal{C}_{b,t,t+\Delta}^{\text{est}|t} + \mathcal{F}_{b,t+\Delta}^{\text{old}|t}}$$

$$1 + s_{b,t_0,T} := \prod_{k=0}^{n-1} \left(1 + \dot{s}_{b,t_k}^{(\Delta)} \right), \quad t_n = T \quad (6)$$

- $\dot{s}_{b,t}^{(\Delta)}$: local quality-adjusted supply change rate for bucket b over $[t, t + \Delta]$
- $s_{b,t_0,T}$: cumulative quality-adjusted supply change for bucket b over $[t_0, T]$
- b : asset bucket under study
- Δ : length of the local measurement period
- $\mathcal{C}_{b,t,t+\Delta}^{\text{real}}$: realized cash flows received over $[t, t + \Delta]$ from all goods in bucket b at $t + \Delta$,
including both pre-existing goods and any newly created goods entering the bucket
- $\mathcal{F}_{b,t+\Delta}^{\text{new}}$: forecast at $t + \Delta$ of cash flows beyond $t + \Delta$ for all goods then in bucket b ,
including both pre-existing goods and any newly created goods entering the bucket
- $\mathcal{C}_{b,t,t+\Delta}^{\text{est}|t}$: cash flows over $[t, t + \Delta]$ expected at t for the original bucket b
- $\mathcal{F}_{b,t+\Delta}^{\text{old}|t}$: forecast formed at t of cash flows beyond $t + \Delta$ for the original bucket b
- t_0 : initial date of the cumulative measurement interval
- T : terminal date of the cumulative measurement interval
- t_k : intermediate chaining dates used to compound local changes

This is an approximation that increases in accuracy as Δ decreases, which will approach practical measurement limitations. The framework still requires robust fundamental/credit analysis models, significantly independent from P_t , to determine $\mathcal{C}_{b,t,t+\Delta}^{\text{est}|t}$, $\mathcal{F}_{b,t+\Delta}^{\text{new}}$, and $\mathcal{F}_{b,t+\Delta}^{\text{old}|t}$, the specifics of which are, again, beyond the scope of this essay. As a sidenote, while it may seem unnatural to claim that changes in cashflow estimations amount to changes in intrinsic asset quality, this isn't inconsistent. It's true that retrospectively, the ultimate cashflows of an asset do turn out to be one certain outcome, and estimations prior to the ultimate outcomes were merely changing predictions of that outcome, and not the actual outcome itself. However, beside the point that practically this distinction is irrelevant because at any one point in time humans interact with investment assets based on their estimated cash flows at that time, the argument can be made that the future is undetermined, and thus the intrinsic quality of an asset at any one point in time is truly equivalent to a probability distribution / estimate of its future cashflows, and that this distribution can evolve in time as new information is gained, without invalidating previous estimations as

errors in quality determination.

2.4 Implementation Groundwork

Tables 1–4 describe starting points for measuring each top level sector’s s_t (Change in quality-adjusted quantity) and respective weights. Figure 1 plots relative top level sector weights for the post war period.

Table 1. General

A	B	C
1	Relative Sector Importance Weight	Change in quality-adjusted quantity
2 Consumer Nondurables:	<i>See Table 2, D2</i>	change in FRED: DNDGRA3Q086SBEA
3 Consumer non-OER Services:	<i>See Table 2, D3</i>	change in: <i>Table 2, E3</i>
4 Consumer Durables:	FRED: K1CTOTL1CD000	change in: BEA Table 8.2. Chain-Type Quantity Indexes for Net Stock of Consumer Durable Goods (line 1)
5 Investment Durables: (residential and nonresidential structures, equipment, Intellectual Property Products)	FRED: K1PTOTL1ES000 (<i>international holders, other than for K1R53101ES000 (residential structures)</i>)	change in: BEA Table 2.2. Chain-Type Quantity Indexes for Net Stock of Private Fixed Assets, Equipment, Structures, and Intellectual Property Products by Type (line 1)
6 Inventories:	BOGZ1LM145020005A	change in: BEA Table 5.8.6B. Real Private Inventories and Real Domestic Final Sales by Industry, Chained Dollars (Line 1: Private Inventories)
7 Share of Franchise Value:	- SUM(<i>Table 3, C2:C5</i>) (<i>international holders</i>)	<i>see section 2.4.7</i>
8 Land:	FRED: BOGZ1FL105010005A+ BOGZ1FL115010005A+ BOGZ1LM155035015A+ BOGZ1LM165035005A- BOGZ1LM155012665A- BOGZ1FL165012665A- BOGZ1FL165013665A (<i>BOGZ1FL165010005A and BOGZ1FL115010005A internationally held, does not include financial firm owned land</i>)	<i>see section 2.4.7</i>
9 Debt:	SUM(<i>Table 4, B2:B9</i>) (<i>upper estimate, international holders</i>)	<i>see section 2.4.7</i>
10 Collectibles/Commodities/Art/Crypto:	?	?
11 Foreign Investment Assets: (investment durables, investment consumables, land, share of future profits, debt)	FRED: IIPUSASSA	change in foreign asset supply according to the same methodology as domestic, with additional weighting by country market share of US spending
12 Government Consumption:	FRED: A955RC1A027NBEA	change in FRED: B822RA3Q086SBEA
13 Government Durables:	FRED: A782RC1A027NBEA	change in FRED: B822RA3Q086SBEA

Table 2. Consumables and Services

A	B	C	D	E
1	step 1	step 2	step 3: resulting stock-comparable construction	Index:
2 Consumer Nondurable Goods:	for each year t - determine durables relative share in CPI-U. Call this: x _t	for each year t - determine nondurable relative share in CPI-U. Call this: y _t	$(y_t/x_t) * K1CTOTL1CD000_t$	<i>see Table 1, C2</i>
3 Consumer non-OER Services:	for each year t - determine durables relative share in CPI-U. Call this: x _t	pre 1983, for each year t - determine share of CPI-U which components are similar to "Owner's Equivalent Rent" and subtract that from the service aggregate's relative share in CPI-U. Post 1983, remove OER component from the services aggregate. Call this new services aggregate: z _t	$(z_t/x_t) * K1CTOTL1CD000_t$	DSERRA3M086SBEA recalculated by excluding OER post 1983 and OER equivalent pre 1983

Table 3. Equity

A	B	C
1	NAV (excluding equity liabilities)	IMA Net Worth (NAV - Equity)
2 nonfinancial corporate	BOGZ1FL102090005A	NCBIMTA027N
3 financial corporate	?	FBNWMTA027N
4 nonfinancial noncorporate	BOGZ1LM112090005A	$(B4 / B2) * C2$
5 financial noncorporate	?	?

Table 4. Debt

A	B
1	Gross Market / Book Value (Stock)
2 Treasury Securities	BOGZ1FL893061105A - BOGZ1FL703061105A (international holders)
3 Agency/GSE backed Securities	BOGZ1FL893061705A - BOGZ1FL703061705A (international holders)
4 Municipal Securities	BOGZ1FL893062005A - BOGZ1FL703062005A (international holders)
5 Corporate and Foreign Bonds	BOGZ1FL893063005A - BOGZ1FL703063005A (international holders)
6 Open market paper	BOGZ1FL883069175A - BOGZ1FL703069175A (domestic holders)
7 Other loans and advances	ASOLAL (upper estimate)
8 Repo	https://www.financialresearch.gov/the-ofr-blog/2025/12/04/sizing-us-repo-market/ ; Q3 2025: 12.6T

Continued on next page

Table 4. Debt (continued)

A	B
1	Gross Market / Book Value (Stock)
9 *NOTE: Mortgages, Consumer credit, Depository institution loans n.e.c. excluded because they involve double counting with because of securitized components, and these are generally held by banks and not available for purchase by nonbanks until securitized. 10 *NOTE: “Other loans and advances” included because it’s a broad catch-all bucket, and some nonbank tradeable/originated loans like private credit / syndicated loans / leveraged loans, but it’s an upper estimate.	

Sources used for tables 1–4: (Federal Reserve Bank of St. Louis, 2026b,q,c,d,r,x,w): (Federal Reserve Bank of St. Louis, 2026m,n,o,p,l,g,h); (Federal Reserve Bank of St. Louis, 2026i,j,k; U.S. Bureau of Economic Analysis, 2026g,i; Federal Reserve Bank of St. Louis, 2026t,u); (Federal Reserve Bank of St. Louis, 2026s,e,f; U.S. Bureau of Economic Analysis, 2026e,b,c,a); (U.S. Bureau of Economic Analysis, 2026d; Federal Reserve Bank of St. Louis, 2026v,a; U.S. Bureau of Economic Analysis, 2025b, n.d., 2025a; Cenicola et al., 2025)

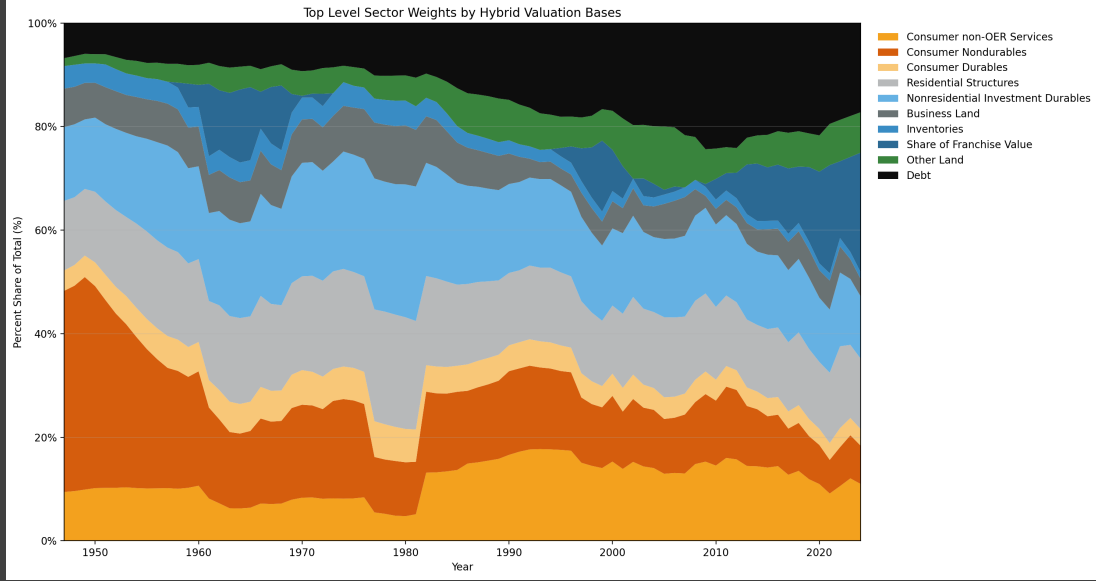


Figure 1. Top-level sector weights by hybrid valuation bases.

Notes: For simplicity, rather than including the foreign investment sector and excluding foreign investment into U.S. goods, typical foreign investable assets—Nonresidential Investment Durables, Business Land, Inventories, SFV, and Debt—were scaled by

$$\frac{\text{Typical Foreign Investable Goods} - \text{Net International Investment Position (IIP)}}{\text{Typical Foreign Investable Goods}}.$$

IIP is not tracked before 1976 (U.S. Bureau of Economic Analysis, 2026f), and is set to zero for these years. Government Expenditure Sector is excluded due to the difficulty of constructing a synthetic Government Consumption stock proxy. Valuation bases are mixed, with preference for market value but including current replacement cost and book value when market value is unavailable. “Open Market Paper” is domestic holders only (Federal Reserve Bank of St. Louis, 2026l), though it should include international investors because it’s adjusted by IIP. The “Other loans and advances” debt category is excluded because it includes an unknown amount of loans not tradeable by the nonbank market (Board of Governors of the Federal Reserve System, 2026b, pp. 34–35). Repo is excluded because data is limited. Financial noncorporate SFV is excluded because of limited data; Nonfinancial Noncorporate SFV is estimated using

$$\left(\frac{\text{Nonfinancial Noncorporate NAV}}{\text{Nonfinancial Corporate NAV}} \right) \times \text{Nonfinancial Corporate SFV}.$$

Financial firms’ land assets are excluded due to double-counting risks with nonfinancial land assets. Sources used for figure: (U.S. Bureau of Economic Analysis, 2026h; Federal Reserve Bank of St. Louis, 2026c,d,r,x,w,m,n,o,p,l,g,h,i,j,k; U.S. Bureau of Economic Analysis, 2026f; U.S. Bureau of Labor Statistics, 2026; U.S. Bureau of Economic Analysis, 2026g,i; Federal Reserve Bank of St. Louis, 2026t,u,s,e,f).

3 Conclusion

Conventional Monetary aggregates are crippled by confusing highly liquid debt assets with money, as well as disregarding internal ledgers, resulting in significant double and

under counting. Conventional inflation measures are woefully ill-equipped to measure general inflation, when considering the full opportunity set of money. In particular, CPI-U fails in its stated goal of tracking changes in the minimum level of expenditure required to maintain a stable utility level, because it doesn't consider the ability to invest and own a home as necessary component of the average urban consumer's utility. Similarly, PCE is the wrong measure of general inflation because it only tracks the orange components of **1**. Even GDP and GDPu deflators are insufficient because they underweight durables (**1**: midtone orange, light gray, and light blue), by only tracking net acquisitions for newly produced durables, whereas the relative importance for those sectors are also influenced by net acquisitions for second hand durables. Additionally, entire top level sector weights sectors for s_t are excluded by GDP/GDPu deflators.:**1** dark gray, dark blue, green, and black. Some of these sectors, particularly *land*, have deep structural limits to supply expansion that diverge from consumer sectors, resulting in significant underestimation of general inflation by conventional measures. Future work should further develop specific components of the theoretical groundwork for alternative metrics proposed in this work, which are designed to address the aforementioned issues.

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